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DECEMBER 23, 1986

**STANDARD FOR
BATTERY CHARGERS**

UL 1236, THIRD EDITION

Accompanying this transmittal notice is a copy of the third edition of UL 1236.

THIS EDITION OF THE STANDARD IS NOW IN EFFECT.

As indicated on the title page, this standard is an American National Standard.

This edition of the standard is intended to update the format and paragraph number system to be consistent with other UL standards.

Revised and/or additional pages may be issued from time to time.



DECEMBER 23, 1986

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UL 1236

**STANDARD FOR
BATTERY CHARGERS**

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THIRD EDITION

December 23, 1986

Approval as an American National Standard covers the numbered paragraphs on pages dated December 23, 1986. These pages should not be discarded when revised or additional pages are issued if it is desired to retain the approved text. Revisions of this standard will be made by issuing revised or additional pages bearing their dates of issue.

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FOREWORD

A. This Standard contains basic requirements for products covered by Underwriters Laboratories Inc. (UL) under its Follow-Up Service for this category within the limitations given below and in the Scope section of this Standard. These requirements are based upon sound engineering principles, research, records of tests and field experience, and an appreciation of the problems of manufacture, installation, and use derived from consultation with and information obtained from manufacturers, users, inspection authorities, and others having specialized experience. They are subject to revision as further experience and investigation may show is necessary or desirable.

B. The observance of the requirements of this Standard by a manufacturer is one of the conditions of the continued coverage of the manufacturer's product.

C. A product which complies with the text of this Standard will not necessarily be judged to comply with the Standard if, when examined and tested, it is found to have other features which impair the level of safety contemplated by these requirements.

D. A product employing materials or having forms of construction differing from those detailed in the requirements of this Standard may be examined and tested according to the intent of the requirements and, if found to be substantially equivalent, may be judged to comply with the Standard.

E. UL, in performing its functions in accordance with its objectives, does not assume or undertake to discharge any responsibility of the manufacturer or any other party. The opinions and findings of UL represent its professional judgment given with due consideration to the necessary limitations of practical operation and state of the art at the time the Standard is processed. UL shall not be responsible to anyone for the use of or reliance upon this Standard by anyone. UL shall not incur any obligation or liability for damages, including consequential damages, arising out of or in connection with the use, interpretation of, or reliance upon this Standard.

F. Many tests required by the Standards of UL are inherently hazardous and adequate safeguards for personnel and property shall be employed in conducting such tests.

GENERAL

1. Scope

- 1.1 These requirements cover battery chargers rated 600 volts or less and intended for household or commercial use to charge rechargeable batteries, in accordance with the National Electrical Code.
- 1.2 These requirements cover portable and stationary battery chargers for indoor and outdoor use and fixed battery chargers for indoor use, for charging automotive or nonautomotive batteries.
- 1.3 A battery charger that is not a complete assembly and depends upon installation in an end product for compliance with the requirements in this standard is investigated under the requirements of this standard and under the standard for the end product.
- 1.4 A battery charger for use with an internal combustion engine driving a centrifugal fire pump is investigated under the requirements of this standard and the applicable requirements for the end product.
- 1.5 These requirements do not cover a battery charger for use in a marina or boatyard, or other marine application excepting a battery charger intended to be permanently installed on a boat; for fire protection signaling service; or an appliance or system in which a battery charger is used.

2. Glossary

2.1 For the purpose of this standard the following definitions apply.

2.2 **AUTOMOTIVE CHARGER** — A battery charger used to charge storage batteries for automobiles, trucks, and other vehicles operated by internal combustion engines, including battery chargers for use in homes and commercial garages. See paragraphs 2.6 and 2.7.

2.3 **CLASS 2 TRANSFORMER** — A step-down transformer having a secondary voltage of not more than 30 volts rms (42.4 volts peak) acceptable for use with Class 2 low-energy supply circuits, in accordance with the National Electrical Code, ANSI/NFPA 70—1984. Unless such a transformer is the energy-limiting type, having sufficient winding impedance to limit the current output to a specified maximum value, an acceptable secondary-circuit or other overcurrent-protective device is included.

2.4 **ENGINE CRANKING** — The operating condition of a battery charger when providing a high-current output for a short time to supplement the battery-output current while cranking an engine. The cranking condition may be designated as engine-start or boost.

2.5 **ENGINE CRANKING SWITCH** — A switch connected in either a primary or secondary circuit to provide a high-current output for a short time.

2.6 **HOUSEHOLD AUTOMOTIVE CHARGER** — A battery charger having an output rating of not more than 10 amperes and 12 volts, used to charge automotive batteries in home garages.

2.7 **FIXED BATTERY CHARGER** — A battery charger that is intended to be permanently connected electrically.

2.8 **HOUSEHOLD NONAUTOMOTIVE CHARGER** — A battery charger for charging:

A. Rechargeable batteries used with home appliances, such as nickel-cadmium types including, but not limited to, Sizes AAA, AA, C, or D; or

B. Storage batteries including battery-packs used with battery-operated home appliances.

2.9 **PLUG-IN CHARGER** — A small household automotive or nonautomotive battery charger for mounting by insertion of its integral attachment-plug blades in a standard wall receptacle.

2.10 PORTABLE BATTERY CHARGER — A cord-connected battery charger that can easily be moved from one place to another for use.

2.11 STATIONARY BATTERY CHARGER — A cord-connected battery charger that is intended to be fastened in place or located in a dedicated place.

3. Components

3.1 A component of a product covered by this standard shall comply with the requirements for that component, and shall be used in accordance with its recognized rating and other limitations of use. A specific requirement for a component need not be applied if an equivalent requirement is included in this standard.

4. Units of Measurement

4.1 If a value for measurement is followed by a value in other units, in parentheses, the second value may be only approximate. The first stated value is the requirement.

CONSTRUCTION

5. Frame and Enclosure

General

5.1 A battery charger shall be formed and assembled so that it will have the strength and rigidity necessary to resist the abuses to which it may be subjected, without increasing the risk of fire, electric shock, or injury to persons due to total or partial collapse with resulting reduction of spacings, loosening or displacement of parts, or other serious defects.

5.2 A battery charger shall be provided with an enclosure acceptable for the application that shall house all live parts other than the power-supply cord or primary connector and the output leads or terminals that may increase the risk of fire, electric shock, or injury to persons under any condition of use.

5.3 A sheet-metal section of the enclosure shall have a thickness not less than that specified in Table 5.1.

Exception: A small part of an enclosure that complies with the requirements in Section 44 need not have the thickness specified in Table 5.1.

**TABLE 5.1
MINIMUM ACCEPTABLE THICKNESSES
OF ENCLOSURE METAL**

Metal	Minimum Thickness, Inch (mm)	
	At Surfaces Other Than Those to Which a Wiring System Is to be Connected in the Field	At Surfaces to Which a Wiring System Is to be Connected in the
Uncoated sheet steel	0.026 (0.66)	0.032 (0.81)
Galvanized sheet steel	0.029 (0.74)	0.034 (0.86)
Nonferrous sheet metal other than copper	0.036 (0.91)	0.045 (1.14)

5.4 An enclosure or part of an enclosure that also serves as a compartment for a rechargeable battery shall be ventilated to permit dispersion of gases from the battery.

Exception: The enclosure for a small, rechargeable battery need not be ventilated. See item A of paragraph 2.8.

5.5 A portable battery charger intended for wall mounting shall employ a keyhole slot or the equivalent as a mounting means.

5.6 A barrier or the equivalent may be used to prevent wall-mounting screws from projecting into a compartment containing electrical parts and reducing spacings to less than that specified in Section 23.

5.7 If an electrical instrument, such as a meter, forms part of the enclosure, the face or the back of the instrument housing, or both together, shall comply with the requirements for an enclosure.

Exception: An electrical instrument connected in a secondary circuit if damage to or deterioration of the materials of which the housing is made will not result in a risk of fire or electric shock.

5.8 Material supporting terminals or used as internal electrical insulation of an electrical instrument shall comply with the requirements in Section 12. Acceptability of spacings between live and dead metal parts connected to the enclosure within the instrument shall be determined by the dielectric voltage-withstand test. See Section 42.

Exception: An electrical instrument connected in a secondary circuit if damage to or deterioration of the materials will not result in a risk of fire or electric shock.

5.9 Supporting feet of a portable battery charger that form part of the enclosure or are needed for ventilation shall be secured and the aging, physical, and flammability properties of the material shall be investigated. The effect of oil and the absorption of moisture on the mounting feet of an automotive battery charger are also to be considered.

Exception: Rubber may be accepted without test if it does not form part of the enclosure and is not needed to provide ventilation. See paragraph 41.5.

5.10 An enclosure, frame, or cover of a material other than sheet metal shall provide protection equivalent to that required by paragraph 5.3 and Table 5.1. For portable battery chargers, see Section 51.

5.11 Among the factors taken into consideration when judging the acceptability of a nonmetallic enclosure are:

- A. Mechanical strength,
- B. Resistance to impact,
- C. Moisture-absorptive properties,
- D. Combustibility,
- E. Resistance to distortion at temperatures to which the material may be subjected under conditions of normal or abnormal usage, and
- F. Resistance to arcing.

The enclosure material is not to display a loss of a property beyond the minimum acceptable level as a result of aging.

5.12 An adhesive used to attach a cover to a battery charger shall be investigated with respect to exposure to environmental conditions, such as high and low temperatures, high and low humidity, and the like, to determine its ability to retain the cover in position.

Exception: Methods utilizing fusion techniques, such as solvent cementing, ultrasonic welding, electromagnetic induction, and thermal welding.

5.13 The combination of enclosure and internal barriers of a battery charger shall reduce the likelihood of molten metal, burning insulation, flaming particles, or the like, falling on combustible materials, including the surface upon which the charger is supported. An internal barrier may be an integral part of a component or of the charger enclosure.

5.14 To comply with the requirement in paragraph 5.13, an enclosure having openings in the bottom shall be provided with a barrier in accordance with paragraph 5.16 and with Figure 5.1 unless the components enclosed are as described in the following:

A. An individually enclosed component if it contains arcing parts, such as a switch, relay, or contactor;

B. A motor without openings in the bottom of the housing unless:

1. The overload protection provided with the motor is such that no burning insulation or molten material falls to the surface that supports the charger when the motor is energized under each of the following fault conditions:

- a. Open main winding,
- b. Open auxiliary winding, and
- c. Starting switch short-circuited;

2. The motor is provided with a thermal motor protector — a protective device that is sensitive to temperature and current — that will prevent the temperature of the motor winding from exceeding:

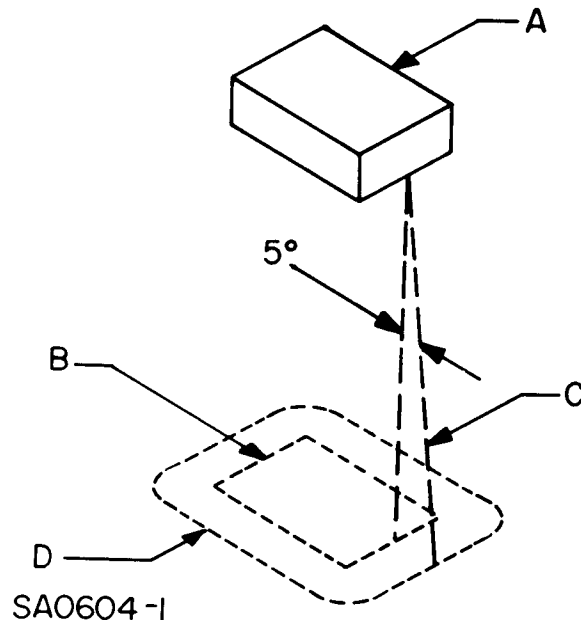
- a. 125°C (257°F) when the motor is running at the maximum load at which it can operate without causing the protector to cycle, and
- b. 150°C (302°F) with the rotor of the motor locked; or

3. The motor complies with the requirements for impedance-protected motors and the temperature of the motor windings will not exceed 150°C during the first 72 hours of operation with the rotor locked;

C. Internal wiring insulated with neoprene, thermoplastic, impregnated asbestos or fiberglass, or an equally fire-retardant material;

D. An individually enclosed fuse, such as an extractor type. Consideration will be given to a fuse enclosed within a transformer winding.

FIGURE 5.1
LOCATION AND EXTENT OF BARRIER



A — Region to be shielded by barrier. This will consist of the entire component if it is not otherwise shielded, and will consist of the unshielded portion of a component partially shielded by the component enclosure or equivalent.

B — Projection of outline of component on horizontal plane.

C — Inclined line that traces out minimum area of barrier. When moving, the line is always (1) tangent to the component, (2) 5 degrees from the vertical, and (3) so oriented that the area traced out on a horizontal plane is maximum.

D — Location (horizontal) and minimum area for barrier. The area is that included inside the line of intersection traced out by the inclined line C and the horizontal plane of the barrier.

5.15 The barrier mentioned in paragraph 5.14 shall be (1) horizontal, (2) located as illustrated in Figure 5.1, and (3) have an area in accordance with that figure.

5.16 If ventilating openings are located in the bottom of the enclosure, a component having a magnetic winding or coil such as a relay or solenoid shall either be:

A. Individually and completely enclosed, or

B. If applicable, subjected to the burnout tests described in paragraphs 49.1—49.3.

5.17 A component of a cart-type automotive battery charger or battery-charger-tester that may produce arcing, such as a snap switch, a relay, or a receptacle, shall be located at least 18 inches (457 mm) above the floor. Other types of automotive battery chargers, or battery-charger-testers for use in commercial garages shall be marked in accordance with paragraph 59.15.

Raintight and Rainproof Enclosures

5.18 The enclosure of a battery charger for outdoor use shall be constructed so as to exclude a beating rain.

5.19 The requirements in paragraphs 5.20—5.25 apply to battery chargers that are intended for permanent outdoor installation.

5.20 When subjected to the water spray test described in Section 43 an enclosure designated as:

A. Raintight shall prevent rain from entering the enclosure.

B. Rainproof shall prevent rain from interfering with the successful operation of the battery charger.

5.21 A raintight or rainproof enclosure shall be marked as specified in paragraph 59.8.

5.22 A gasket employed in a battery charger, including a cart-type automotive charger as specified in the exception to paragraph 59.19, to make an enclosure raintight or rainproof shall be tested as specified in Section 53.

5.23 A raintight or rainproof enclosure shall be provided with external means for mounting.

Exception: A rainproof enclosure may be provided with internal means for mounting if the mounting means is constructed so as to prevent water from entering the enclosure.

5.24 For a raintight enclosure, all openings for conduit other than in the bottom shall be threaded.

5.25 For a rainproof enclosure, openings for conduit shall be threaded unless they are located wholly below the lowest terminal lug or other live part within the enclosure, and there shall be provision for drainage of the enclosure if knockouts or unthreaded holes are provided other than in the bottom.

6. Accessibility of Live Parts

6.1 An opening in the enclosure of a battery charger is acceptable if a probe as illustrated in Figure 6.1 when inserted through the opening cannot be made to touch any uninsulated live part that can cause electric shock.

Exception: As indicated in paragraph 6.2.

6.2 There shall be no openings in the enclosure of a household nonautomotive battery charger except for the input and output cord or terminals.

Exception No. 1: A battery charger intended to charge storage batteries for use with gardening appliances, lawn mowers, and tools may employ ventilating openings complying with the requirements in paragraph 6.1.

Exception No. 2: Ventilating openings may be employed (1) in the bottom of the enclosure of other than a plug-in battery charger, if the construction includes baffles or the equivalent to prevent emission of flame, molten metal, or burning insulation, and (2) in the top or sides of the enclosure if the construction is such that small objects, such as hairpins and paper clips cannot contact uninsulated live parts that can cause electric shock.

6.3 Glass or thermoplastic covering an opening for user servicing, such as replacing a pilot lamp, and that reduces the risk of unintentional contact with a live part involving a risk of electric shock shall be:

A. A material complying with the (1) impact test specified in Section 44, (2) abnormal tests specified in Section 47, and (3) flammability tests for thermoplastics as specified in Section 51; and

B. Reliably retained in place.

6.4 An uninsulated live part that can cause electric shock shall be located or enclosed so that protection against unintentional contact is provided.

6.5 A door or cover that provides access to a live part that can cause electric shock shall be securely held in place so that it can be opened or removed only by using a tool.

Exception: A door or cover that provides access to a live part that does not involve a risk of electric shock shall be securely held in place, but need not be secured so that it is necessary to use a tool to open or remove it.

6.6 A household nonautomotive battery charger may include a hinged or sliding cover or tray, or similar construction, if it gives access only to uninsulated battery terminals or clips, see item A of paragraph 2.8.

6.7 The door or cover of an enclosure shall be hinged or attached in an equivalent manner if it provides access to an overload-protective device, the functioning of which requires renewal, or if it is necessary to open the cover in connection with the operation of the protective device. A door or cover providing access to a fuseholder shall be tight-fitting and shall be positively held closed.

6.8 The operating handle of a circuit breaker, operating button of a manually operable protector, the capped portion of an extractor-type fuseholder, or a similar part may project outside the enclosure.

7. Assembly

7.1 An uninsulated live part shall be secured to the base or surface so that it will be prevented from rotating or shifting in position as the result of normal stresses if such movement might result in a reduction of spacings below the minimum acceptable values. See paragraphs 23.1 and 23.2.

7.2 A component such as a rectifier element, control switch, lampholder, attachment plug receptacle, or plug connector shall be mounted securely and shall be prevented from turning by means other than friction between surfaces.

Exception No. 1: A switch need not be mounted as described if the following conditions are met:

A. The switch is a plunger or other type that does not tend to rotate when operated. A toggle switch is considered to be subject to forces that tend to turn the switch;

B. The means for mounting the switch makes it unlikely that operation of the switch will loosen it;

C. Spacings are not reduced below the minimum acceptable values if the switch rotates; and

D. Operation of the switch is by mechanical means rather than by direct contact by persons.

Exception No. 2: A lampholder of the type in which the lamp cannot be replaced, such as a sealed neon pilot or indicator light, if rotation cannot reduce spacings below the minimum acceptable value.

7.3 A small stem-mounted device having a single-hole mounting means may be prevented from rotating by a properly applied lock washer.

8. Protection Against Corrosion

8.1 Iron and steel parts shall be protected against corrosion by enameling, painting, galvanizing, plating, or an equivalent means.

Exception No. 1: Bearings, laminations, other parts of iron or steel such as washers and screws need not be protected from corrosion.

Exception No. 2: A part need not be protected from corrosion if corrosion of the part would not result in a risk of fire, electric shock, or injury to persons.

8.2 The interior of a metal compartment housing a lead-acid battery shall be protected by two coats of acid-resistant paint, two coats of enamel individually baked on, or the equivalent.

8.3 An enclosure of sheet-steel intended for outdoor use shall be protected against corrosion by one of the following coatings:

A. Hot-dipped mill-galvanized sheet steel conforming with the coating Designation G90 in Table I of the Standard Specification for Sheet Steel Zinc Coated (Galvanized) by Hot-Dip Process, General Requirements, ASTM A525—80a, with not less than 40 percent of the zinc on any side, based on the minimum single-spot test requirement in this ASTM designation. The weight of the zinc coating may be determined by any acceptable method; however, in case of question, the weight of the coating shall be established in accordance with the Test Method of ASTM Designation A90—69.

B. A zinc coating, other than that provided on hot-dipped mill-galvanized sheet steel, uniformly applied to an average thickness of not less than 0.00061 inch (0.0155 mm) on each surface with a minimum thickness of 0.00054 inch (0.0137 mm). The thickness of coating shall be established by the metallic-coating-thickness test described in Section 54. An annealed coating shall comply with paragraphs 8.8 and 8.9.

C. A zinc coating conforming with item 1 or 2 below and with one coat of an organic finish of the epoxy or alkyd-resin type or other outdoor paint on each surface, applied after forming. The acceptability of a paint can be determined by consideration of its composition or by corrosion tests if these are considered necessary.

1. Hot-dipped mill-galvanized sheet steel conforming with the coating Designation G60 or A60 in Table I of the Standard Specification for Sheet Steel, Zinc-Coated (Galvanized) by Hot-Dip Process, General Requirements, ASTM A525—80a, with not less than 40 percent of the zinc on any side, based on the minimum single-spot test requirement in this ASTM designation. The weight of zinc coating may be determined by any acceptable method; however, in case of question the weight of coating shall be established in accordance with the Test Method of ASTM Designation A90—69. An A60 alloyed coating shall comply with paragraph 8.5.

2. A zinc coating, other than that provided on hot-dipped mill-galvanized sheet steel, uniformly applied to an average thickness of not less than 0.00041 inch (0.0104 mm) on each surface with a minimum thickness of 0.00034 inch (0.0086 mm). The thickness of the coating shall be established in accordance with the metallic-coating-thickness test described in Section 54. An annealed coating shall comply with paragraph 8.5.

D. A cadmium coating not less than 0.0010 inch (0.025 mm) thick on both surfaces. The thickness of coating shall be established in accordance with the metallic-coating-thickness test described in Section 54.

E. A cadmium coating not less than 0.00075 inch (0.0191 mm) thick on both surfaces with one coat of outdoor paint on both surfaces; or not less than 0.00051 inch (0.0130 mm) thick on both surfaces with two coats of outdoor paint on both surfaces. The thickness of the cadmium coating shall be established in accordance with the metallic-coating-thickness test described in Section 54 and the paint shall be as specified in item C.

F. Other finishes, including paints, metallic finishes, or combinations of the two may be accepted when comparative tests with galvanized sheet steel — without annealing, wiping, or other surface treatment — conforming with item A indicate they provide equivalent protection. Among the factors that are taken into consideration when judging the acceptability of such coating systems are exposure to salt spray, moist carbon dioxide-sulfur dioxide-air mixtures, moist hydrogen sulfide-air mixtures, ultraviolet light, and water.

8.4 A sheet-steel enclosure for a cart-type automotive battery charger as specified in the exception to paragraph 59.19 shall be protected against corrosion by one of the coatings specified in paragraph 8.3 or by a hot-dipped mill-galvanized sheet-steel coating conforming with the coating Designation G60 in Table I of ASTM Designation A525—80a, with not less than 40 percent of the zinc on any side, based on the minimum single-spot test requirement in this ASTM designation. The weight of the zinc coating may be determined by any acceptable method; however, in case of question, the weight of the coating shall be established in accordance with the Test Method of ASTM Designation A90—69.

8.5 An annealed coating on sheet steel that is bent or similarly formed, or extruded or rolled at the edge of holes after annealing shall be additionally painted in the affected area if the process damages the zinc coating.

8.6 If flaking or cracking of the zinc coating at the outside radius of the bent or formed section is visible at 25 power magnification, the zinc coating is considered to be damaged. Simple sheared or cut edges and punched holes are not required to be additionally protected, but extruded and rolled edges at holes are to comply with the requirements in paragraph 8.5.

8.7 The acceptability of outdoor paint may be determined by consideration of its composition or, if necessary, by tests.

9. Supply and Output Connections

Permanently Connected Battery Chargers

General

9.1 A fixed battery charger shall have provision for the connection of one of the wiring systems that is acceptable for the application in accordance with the National Electrical Code.

9.2 A knockout in a sheet-metal enclosure shall be secured and shall be removable without undue deformation of the enclosure.

9.3 A knockout shall be surrounded by a flat surface acceptable for seating a conduit bushing or locknut of the appropriate size.

9.4 A terminal box or compartment in which field-wiring connections are to be made shall be located so that the connections may be readily inspected after the battery charger is installed as intended.

9.5 A field-wiring compartment intended for connection of a wiring system shall be attached to the battery charger so that it will be prevented from turning.

9.6 If a separate field-wiring compartment is not provided, space for field-wiring connections shall be located near the connecting means and shall permit routing of the wiring away from live parts and rough metal. See paragraph 9.4.

Wiring Terminals and Leads

9.7 A field-wiring terminal is a terminal to which power-supply, control, output, or other permanent connections will be made in the field when the charger is installed.

9.8 Field-wiring terminals or leads shall be acceptable for the connection of conductors having an ampacity appropriate for the rating of the charger.

9.9 A wiring terminal shall be provided with a soldering lug or pressure terminal connector securely fastened in place — for example, firmly bolted or held by a screw.

Exception: A wire-binding screw may be employed at a wiring terminal intended for connection of a No. 10 AWG (5.3 mm²) or smaller conductor if up-turned lugs or the equivalent are provided to hold the wire in position.

9.10 A wiring terminal shall be provided with a means other than friction between surfaces to reduce the likelihood of turning or shifting in position. This may be accomplished by two screws or rivets; by square shoulders or mortises; by a dowel pin, lug, or offset; by a connecting strap or clip fitted into an adjacent part; or by an equivalent method.

9.11 A wire-binding screw at a field-wiring terminal shall not be smaller than No. 10 (4.8 mm diameter).

Exception No. 1: A No. 8 (4.2 mm diameter) screw may be used at a terminal intended only for the connection of a No. 14 AWG (0.82 mm²) conductor.

Exception No. 2: A No. 6 (3.5 mm diameter) screw may be used for the connection of a No. 16 AWG (1.3 mm²) or No. 18 AWG (0.82 mm²) control-circuit conductor.

9.12 A wire-binding screw shall thread into metal.

9.13 A terminal plate tapped for a wire-binding screw shall be of metal not less than 0.050 inch (1.27 mm) thick.

Exception: A plate not less than 0.030 inch (0.76 mm) thick may be used if the tapped threads have acceptable mechanical strength.

9.14 There shall be two or more full threads in the metal of a terminal plate. The metal may be extruded at the tapped hole to provide at least two full threads.

9.15 Upturned lugs, a cupped washer, or the equivalent shall be capable of retaining a conductor of the size required by paragraph 9.11, but not smaller than No. 14 AWG (2.1 mm²), under the head of the screw.

9.16 The free length of a lead inside an outlet box or wiring compartment shall be 6 inches (152 mm) or more if the lead is intended for field connection to an external circuit.

Exception: A lead may be less than 6 inches long if it is evident that the use of a longer lead may result in a risk of fire or electric shock.

Identification

9.17 A fixed battery charger rated 125 or 125/250 volts (3-wire) or less, and employing a lampholder of the Edison screw-shell type, or a single-pole switch or overcurrent-protective device other than an automatic control without a marked off position shall have one terminal or lead identified for the connection of the grounded conductor of the supply circuit. A single-pole switch or single-pole overcurrent-protective device, other than an automatic control without a marked off position, shall not be connected to the terminal or lead intended for connection to the grounded conductor of the supply circuit.

9.18 A field-wiring terminal intended for connection of a grounded circuit conductor shall be made of or plated with metal substantially white in color and shall be readily distinguishable from the other terminals; or proper identification of that terminal shall be clearly shown in some other manner, such as on an attached wiring diagram.

9.19 The surface of a lead for the connection of a grounded circuit conductor shall be white or natural grey and shall be readily distinguishable from other leads.

9.20 A battery charger shall have a terminal or lead for connecting the metal enclosure and enclosure parts to ground.

Cord-Connected Battery Chargers

General

9.21 A portable or stationary battery charger shall be provided with a flexible cord in accordance with Table 9.1 and an attachment plug for connection to the power supply. The length of cord external to the battery charger and including the attachment plug shall not be less than 6 feet (1.8 m) as measured from the face of the attachment plug to the point of attachment or entry.

Exception: A nonautomotive battery charger weighing 1 pound (454 g) or less, is acceptable if the total length of the input and output cords is 6 feet or more and the length of the input cord is at least 3 feet (914 mm).

TABLE 9.1
ACCEPTABLE FLEXIBLE CORDS FOR
BATTERY CHARGERS

Type of Charger	Type of Cord
Household, Automotive ^a and Nonautomotive ^a	SP-2, SPT-2, SV, SVO, SVT, or SVTO
Other Than Household For use where oil is a consideration	SJT ^b , ST ^b , SJO ^c , SO ^c , SKTP ^c , STO ^c
For use where oil is not a consideration	SJT ^b , ST ^b , SJO ^c , SO ^c ,

^aOutput ratings not exceeding 10 amperes and 12 volts.

^bShall be marked "Oil-Resistant."

^cShall be marked "W-A" when used on a battery charger intended to be used outdoors.

9.22 A fixed battery charger may require a supply connection that will facilitate the interchange of equipment to maintain continuous service or otherwise meet conditions of use. For such service, a Type S flexible cord with grounding conductor may be employed and may be of a length appropriate for the purpose. Normally a 24 inch (610-mm) length of cord will be sufficient for the plug and receptacle connection.

9.23 The current and voltage ratings of a flexible cord and attachment plug for a battery charger having a current input rated:

A. 15 amperes or less shall not be less than the input rating of the charger.

B. More than 15 amperes shall not be less than the input rating of the charger except that if the charger is intended to be continuously loaded for 3 hours or more, the ampacity of the attachment plug shall not be less than 125 percent of the input rating.

9.24 The ampacity of the flexible cord for a battery charger having an engine-cranking circuit switch and marked with a duty cycle — see paragraph 2.5 — is to be calculated by using the following equation:

$$I = \sqrt{I_1^2 d + I_2^2 (1-d)}$$

In which:

I is the ampacity of the cord as given in the National Electrical Code, ANSI/NFPA 70-1984.

I_1 is the input engine-cranking current,

I_2 is the input current measured while maximum continuous-duty output current is being delivered, and

d is the rated duty cycle expressed as a fraction of total time.

9.25 A flexible cord incorporating a pendant switch connected in a line-voltage circuit shall be equivalent to the power-supply cord.

9.26 In a portable or stationary battery charger rated 125 or 125/250 volts (3-wire) or less, a switch or overcurrent-protective device of the single-pole type, other than an automatic control without a marked off position, shall be connected in a circuit to the cord conductor not intended to be grounded.

Output Connections

9.27 A cord-connected or plug-in battery charger shall be provided with (1) an integral output cord or leads terminating in a battery connector, or (2) an integral output receptacle directly attached to the charger enclosure and a cord set.

Exception: A nonautomotive battery charger having Class 2 output characteristics may employ terminals for output connections.

9.28 The output cord or leads required by paragraph 9.27 for other than a plug-in automotive charger, whether permanently attached to the battery charger or provided as a separate cord set, shall not be less than 6 feet (1.83 m) long measured from the point of exit. The output cord of a plug-in automotive charger shall not be less than 12 feet (3.7 m) long.

Exception: The output cord of a commercial nonautomotive charger for which a shorter length of cord or leads are utilized to meet conditions of use shall not be less than 2 feet (609 mm) long. The manufacturer's instructions provided with a charger intended to charge other than sealed rechargeable cells shall specify that the batteries are to be placed as far away from the charger as the output leads will permit during charging.

9.29 The output cord used with a battery charger, other than the automotive or household nonautomotive type having Class 2 output characteristics, shall be as specified in Table 9.1 or shall be of a type equally serviceable for the application. The use of individual output leads will be considered on the basis of equivalent insulation thickness and acceptability for other conditions of service as specified in paragraph 11.1.

9.30 The output wiring of an automotive battery charger shall be either (1) a flexible cord equivalent to the power-supply cord, or (2) individual conductors that comply with the requirements in paragraph 9.38.

9.31 The output wiring of a household nonautomotive battery charger that complies with the requirements in paragraph 9.40 may be parallel cord insulated with rubber, neoprene, or thermoplastic having a wall thickness not less than 0.013 inch (0.33 mm).

9.32 An automotive battery charger intended for connection to an automobile cigar lighter receptacle shall be provided with a plug that complies with the requirements in the Standard for Attachment Plugs and Receptacles, UL 498. The connector plug shall incorporate a fuse or other protective device having a voltage and current rating sized for the circuit in which it is intended to be connected.

Exception: The protective device may be provided in the output cord not more than 5 inches (127 mm) from the connector plug.

9.33 With reference to paragraph 9.32, if a fuse is employed, the fuse type and size shall be marked on the fuseholder, as described in paragraph 59.27.

9.34 An output connection of a household battery charger, such as a connector, terminal, or clip, shall not be exposed if a risk of electric shock may result.

Exception: A permanently connected battery charger to which output leads are to be field-connected to terminals inside an enclosed wiring compartment.

9.35 A commercial battery charger having output voltage greater than 42.4 volts peak and intended for charging several batteries connected in series shall have connector guards secured to reduce the likelihood of exposure of a connector more than is necessary to make connection to a battery terminal, and shall have a cautionary marking in accordance with paragraph 59.17.

Exception: As provided in the exception to paragraph 9.34.

9.36 The output leads and connectors of a battery charger, other than a nonautomotive household-type, shall be of adequate size and have insulation thick enough to withstand the mechanical abuses to which they are subjected. The insulation of the output connectors on a battery charger with an output voltage of 42.4 volts peak or less shall be reliably secured and substantially cover the connector. A battery charger having an output voltage greater than 42.4 volts peak, designed for charging more than two automotive-type batteries connected in series, shall employ closed-loop lugs or the equivalent for connection to the battery terminals.

Exception: The portion of a connector that attaches to battery terminals need not be covered.

9.37 A spring clip is not considered to be equivalent to a closed-loop lug unless provided with an insulator secured so that grounding or short-circuiting is not likely to result if the connector is displaced.

9.38 With reference to paragraph 9.36:

A. The wall thickness of the insulation on the leads shall not be less than:

1. For an automotive charger not exceeding 42.4 volts output, 1/32-inch (0.8-mm) thermoplastic, 3/64-inch (1.2-mm) rubber or neoprene of the type specified in paragraph 11.1, or 1/16-inch (1.6-mm) rubber or neoprene.

2. For an automotive charger exceeding 42.4 volts output, 1/16-inch thermoplastic, or 5/64-inch (2.0-mm) rubber or neoprene of the type specified in paragraph 11.1.

B. The wall thickness of insulation on a connector guard shall not be less than:

1. For an automotive charger not exceeding 42.4 volts output, 1/32-inch thermoplastic, or 1/16-inch rubber or neoprene.

2. For an automotive charger exceeding 42.4 volts output, 1/16-inch thermoplastic or 5/64-inch rubber or neoprene.

9.39 Acceptability of the material of a connector guard is to be determined for a battery charger having output exceeding 42.4 volts. A connector guard of rubber, neoprene, polyvinyl chloride, or a copolymer thereof shall be tested in accordance with Tests of Connector Guards, Section 52. A connector guard of insulating material other than those specified in Section 52 shall be investigated in accordance with the requirements in paragraph 12.2.

9.40 A household nonautomotive battery charger shall have output connections supplied from the secondary of a Class 2 transformer or shall have an impedance in the secondary circuit to provide Class 2 characteristics at the output connections.

Exception: A household nonautomotive battery charger need not comply if:

A. An interlock switch is provided within the charger enclosure to de-energize accessible live parts such as battery terminals or clips, or

B. An output cord equivalent to that specified in Table 9.1 is provided and terminated by means of an insulated connector having recessed live parts.

9.41 With reference to paragraph 9.40, the output will be considered as having Class 2 characteristics if the open-circuit voltage does not exceed 42.4 volts peak and, with the output terminals short-circuited for 1 minute, the current does not exceed 8 amperes.

9.42 A metal enclosure of a battery charger shall not be connected to an electrical circuit of the charger.

Strain Relief

9.43 Strain relief shall be provided to prevent a pull exerted on the supply or output cord from being transmitted to terminals, splices, or interior wiring.

9.44 A metal strain-relief clamp or band may be used without supplementary protection on a Type SV, SVO, SJ, SJO, S, SO, SJT, SJTO, ST, or STO cord. A metal strain-relief clamp or band may be used on a Type SP-2, SPT-2, SVT, or SVTO cord only if supplementary nonconductive, mechanical protection is provided over the cord.

9.45 If a knot in a flexible cord serves as strain relief, the surfaces that the knot may touch shall be free from burrs, fins, sharp edges, and projections that may damage the cord.

9.46 Means shall be provided to prevent the flexible cord from being pushed into the battery charger enclosure through the cord-entry hole if such displacement may (1) result in mechanical damage to the cord, (2) expose the cord to a temperature higher than that for which it is rated, or (3) reduce spacings such as to a metal strain-relief clamp, below the minimum acceptable values.

Bushings

9.47 A bushing or the equivalent shall be provided at a point where a flexible cord passes through an opening in a wall, barrier, or enclosing case. The bushing shall be substantial, reliably secured in place, and shall have a smooth, rounded surface against which the cord may bear. If Type SP-2 or SPT-2 cord is employed, the wall or barrier is of metal, and the construction is such that the cord may be subjected to stress or motion, the bushing shall be an insulating bushing.

Exception: For a cord hole in wood, porcelain, phenolic composition, or other acceptable nonconductive materials, a smooth, rounded surface is considered to be the equivalent of an insulating bushing.

9.48 Ceramic materials and some molded compositions may be used for insulating bushings.

9.49 Vulcanized fiber may be employed if the bushing is not less than 3/64 inch (1.2 mm) thick, and if formed and secured in place so that it will not be adversely affected by conditions of ordinary moisture.

9.50 A separate soft-rubber, neoprene, or polyvinyl chloride bushing may be used in the frame of a motor or in the enclosure of a capacitor attached to a motor, if the bushing is:

A. Not less than 3/64 inch (1.2 mm) thick; and

B. Located so that it will not be exposed to oil, grease, oily vapor, or other substances having a deleterious effect on the compound employed.

9.51 A bushing of a material specified in paragraph 9.50 may be employed at any point in a battery charger if used in conjunction with a type of cord for which an insulating bushing is not required.

9.52 If a bushing of a material mentioned in paragraph 9.50 is used, the hole in which the bushing is mounted is to be smooth and free from sharp edges.

9.53 An insulating bushing molded integrally with the supply cord is acceptable on a Type SP-2 or heavier cord provided the built-up section is not less than 1/16 inch (1.6 mm) thick where the cord passes through the enclosure.

9.54 An insulated metal grommet is acceptable as an insulating bushing if the insulating material is not less than 1/32 inch (0.8 mm) thick and completely fills the space between the grommet and the metal in which it is mounted.

10. Live Parts

10.1 A current-carrying part shall be of silver, copper, a copper alloy, or other material acceptable for the application.

10.2 Plated iron or steel may be used for a current-carrying part (1) if acceptable in accordance with paragraph 3.1, (2) within a motor, and (3) in a secondary circuit rated 42.4 volts peak — 30 volts rms — or less; but unplated iron or steel is not acceptable. Stainless steel and other corrosion-resistant alloys may be used for current-carrying parts.

10.3 An uninsulated live part shall be secured to an insulating base or mounting surface so that it will be prevented from turning or shifting in position if such motion might result in a reduction of spacings below the minimum acceptable values.

10.4 Friction between surfaces is not acceptable as a means to prevent shifting or turning of a live part, but a properly applied lock washer is acceptable.

11. Internal Wiring

General

11.1 The internal wiring of a battery charger shall consist of wire of a type that is acceptable for the application when considered with respect to (1) exposure to oil, grease, or other substances that may have a deleterious effect on the insulation, and (2) the temperature and voltage, and (3) other conditions of service to which the wiring may be subjected.

Exception: Internal wiring in a secondary circuit of 50 volts open circuit or less if it is acceptable when judged as bare wire with respect to intermingling with the primary circuit wiring and separation from uninsulated live parts and grounded metal.

11.2 The thickness of insulation on wiring shall not be less than 3/64 inch (1.2 mm) for unbraided neoprene-insulated wire, 1/32 inch (0.8 mm) for thermoplastic-insulated wire, and 1/32 inch for rubber-insulated wire with an impregnated braid.

Exception: As provided in the exception to paragraph 11.1.

11.3 Unless a charger includes a resistor or other heat-producing component and the wire is subjected to a temperature of more than 80°C (176°F), or other need is demonstrated, Type AF or CF wire shall not be employed. If used, these types shall not be exposed to moisture, including condensation resulting from operation of the charger.

Exception: As provided in the exception to paragraph 11.1.

11.4 The length of the power-supply cord inside a battery charger shall be not more than that needed for electrical connections.

11.5 The jacket of a flexible cord inside the enclosure shall not be stripped to expose the individual conductors unless:

- A. The insulation on the individual conductors is equivalent to that required by paragraph 11.2;
- B. The individual conductors are supported in a manner positively separating them from live parts and dead metal parts; or
- C. Supplementary insulation equivalent to that required by paragraph 11.2 is reliably secured on each individual conductor.

Tubing

11.6 Insulation of internal wiring consisting of coated fabric, thermoplastic, or other types of tubing is to be considered with respect to electrical, mechanical, and flammability properties of the material.

Protection of Wiring

11.7 Internal wiring shall be protected if, when judged in accordance with paragraph 6.1, it is accessible.

Exception: Internal wiring need not be protected if it is located and secured within the enclosure so that it is not likely to be subjected to stress or mechanical damage.

11.8 Wires within an enclosure, compartment, raceway, or the like shall be located or protected to reduce the likelihood of contact with any sharp edge, burr, fin, moving part, or the like that can damage the conductor insulation.

11.9 A hole in a sheet-metal wall within the overall enclosure of a charger through which insulated wires pass shall be provided with smooth, rounded surfaces upon which the wires may bear, to reduce the likelihood of abrasion of the insulation.

Electrical Connections

11.10 Aluminum conductors, insulated or uninsulated, used as internal wiring, such as for interconnection between current-carrying parts or in a component winding, shall be terminated at each end by a method acceptable for the combination of metals involved at the points of connection.

11.11 With reference to paragraph 11.10, a wire-binding screw or a pressure wire connector used as a terminating device shall be acceptable for use with aluminum under the conditions involved — for example, temperature, heat cycling, and vibration.

11.12 A splice or connection shall be mechanically secure and shall make reliable electrical contact.

11.13 A soldered connection shall be mechanically secured before being soldered.

11.14 A splice shall be provided with insulation equivalent to that of the wires involved unless permanent spacings will be maintained between the splice and other metal parts.

11.15 If the voltage involved is 250 volts or less, insulation consisting of two layers of thermoplastic tape, of two layers of friction tape, of one layer of friction tape and one layer of rubber tape, that has been investigated and found acceptable for the purpose, is acceptable on a splice.

11.16 In determining if splice insulation consisting of coated fabric, thermoplastic, or other tubing is acceptable, consideration is to be given to such factors as its electrical and mechanical properties and its flammability. Thermoplastic tape wrapped over a sharp edge is not acceptable.

11.17 The means of connecting stranded internal wiring to a wire-binding screw shall be such that loose strands of wire will be prevented from contacting other live parts not always of the same polarity as the wire, and from contacting dead metal parts. This may be accomplished by the use of a pressure terminal connector, a soldering lug, a crimped eyelet, soldering of all strands together, or other reliable means.

Separation of Circuits

11.18 Unless provided with insulation acceptable for the highest of the circuit voltages, insulated conductors of different circuits within a battery charger, including wires in a terminal box or compartment, shall be separated by barriers or shall be segregated and shall be so separated or segregated from uninsulated live parts connected to different circuits.

11.19 Segregation of insulated conductors may be accomplished by clamping, routing, or an equivalent means that will maintain permanent separation from insulated and uninsulated live parts and from conductors of a different circuit.

Barriers

11.20 A barrier used to separate or segregate internal wiring shall have adequate mechanical strength, and shall be reliably held in place.

11.21 A barrier of metal shall be at least as thick as specified in paragraph 5.3. A barrier of insulating material shall not be less than 0.028 inch (0.71 mm) thick and shall be supported so that it cannot be readily deformed to defeat its purpose.

12. Insulating Materials

12.1 An insulating washer, a bushing, or the like and a base or a support for mounting a live part shall be of moisture-resistant material that will not be affected adversely by the temperatures and stresses to which it will be subjected under conditions of use.

12.2 Insulating material is to be judged with respect to its acceptability for the application. Materials such as mica, ceramic, and some molded compounds are usually acceptable for the sole support of live parts. If an investigation is necessary to determine whether a material is acceptable, consideration is to be given to (1) its mechanical strength, resistance to ignition, dielectric strength, insulation resistance, and heat-resistance qualities in both the aged and unaged conditions; (2) the degree to which it is enclosed; and (3) any other feature affecting the risk of fire, electric shock, or injury to persons. All factors are to be considered with respect to conditions of service.

13. Motors

13.1 A fan motor used in a battery charger shall be provided with either of the following:

A. Thermal protection complying with the requirements in the Standard for Thermal Protectors for Motors, UL 547; or

B. Impedance protection complying with the Standard for Impedance Protected Motors, UL 519.

Exception: A motor complying with (1) the requirements in item B of paragraph 5.14 and (2) the abnormal test required by paragraph 47.10 need not comply with item A or B.

13.2 A motor winding shall resist the absorption of moisture.

Exception: For household indoor nonautomotive battery chargers and totally enclosed plug-in chargers, film-coated wire need not be additionally treated to retard the absorption of water. Cloth coil wrap, fiber slot liners, and similar moisture-absorptive materials are to be impregnated or otherwise treated to retard the absorption of water.

14. Transformers

General

14.1 A transformer shall have its primary winding electrically isolated from its secondary winding and shall be constructed as specified in paragraphs 14.2—14.11 so that there is no electrical connection — under normal and overload conditions — between the primary and secondary windings, between the primary winding and the core, or between separate adjacent secondary windings if such a connection might result in a risk of fire or electric shock.

Exception: The requirements in paragraphs 14.4—14.11 do not apply to control-circuit transformers used only for controlling internal loads within the battery charger such as pilot lights, fans, contactor coils, relays, and timers.

14.2 A transformer coil shall be treated with an insulating varnish and baked, or otherwise impregnated to exclude moisture or acid vapor.

Exception: For a household indoor nonautomotive battery charger and a totally enclosed plug-in charger, film-coated wire need not be additionally treated. Cloth coil wrap and similar moisture-absorptive materials are to be impregnated or otherwise treated to retard the absorption of water.

14.3 With reference to paragraph 14.2, the requirement concerning acid vapor applies only to a battery charger with a self-contained lead-acid-cell battery.

Insulation

14.4 A transformer shall be constructed to provide insulation required for the transformer temperature class between:

- A. The primary wires of opposite polarity,
- B. The primary winding and any secondary winding,
- C. The primary winding and the core,
- D. A secondary winding having a potential of more than 30 volts rms (42.4 volts peak) and the core,
- E. The primary winding or lead connections and the transformer enclosure or other dead metal parts,
- F. The primary-lead connections and the adjacent winding,
- G. The secondary-lead connections of a coil having a potential greater than 30 volts rms (42.4 volts peak) and the adjacent winding, and

H. The secondary-lead connections and the primary winding.

Exception: Insulation is not required where the spacing specified in Table 23.4 is provided.

14.5 The transformer windings mentioned in paragraph 14.4 include the start, all taps, finish, and crossover leads up to the point where insulated leads are provided.

14.6 The transformer insulation shall be as specified in Table 14.1 as applicable.

14.7 Insulation between any winding or lead connection and dead metal parts shall be electrical grade paper, waxed or otherwise treated to resist the absorption of moisture:

- A. Having a total thickness of not less than 0.013 inch (0.33 mm) if used in conjunction with an air spacing of one-half that specified in Table 23.4; or
- B. Having a total thickness of not less than 0.028 inch (0.71 mm) if the insulation is in contact with the enclosure.

Exception No. 1: A material having a thickness less than that specified in items A and B may be used provided that it is mechanically and thermally equivalent to items A and B and that the product of volts per mil and the material thickness is not less than 2500 volts for item A and 5000 volts for item B.

Exception No. 2: Where the spacing specified in Table 23.4 is provided, the insulation thickness in item A or B need not be provided.

TABLE 14.1
TRANSFORMER INSULATION

Location of Insulation	Acceptable Types of Insulation		
Insulation between the primary wires of opposite polarity and between secondary wires of opposite polarity having a potential greater than 30 volts, rms (42.4 volts peak) ^a	A	B	C
Insulation between the primary and any secondary winding ^{a,b}	A	B	C
Insulation between any winding having a potential greater than 30 volts (42.4 volts peak) and the core ^a	A	B	C

A. Electrical grade paper that is waxed or otherwise treated to retard the absorption of moisture and that has a total thickness of not less than 0.028 inch (0.71 mm).

B. A thermoplastic or thermoset coil form not less than 0.028 inch thick.

C. Material having a thickness less than 0.028 inch may be used provided that it is mechanically, electrically, and thermally equivalent to item A or B and that the product of volts per mil and the material thickness is not less than 500 volts.

^aInsulation is not specified where the spacing specified in Table 23.4 is provided.

^bInsulation is not specified between primary and secondary windings that are separated by a grounded copper shield or the equivalent having a current-carrying capacity of not less than the primary winding. If conductors having a circular cross-section are used, two layers shall be employed to reduce air gaps and completely isolate the windings. Following the normal and abnormal operation tests, the insulation between the windings and the shield shall withstand the dielectric voltage-withstand test described in paragraph 42.1.

14.8 Insulation between the crossover leads and (1) the turns of a different winding, (2) the metal battery charger enclosure, or (3) the core shall be one of the following:

A. Electrical grade paper, waxed or otherwise treated to resist the absorption of moisture, having a total thickness of not less than 0.028 inch (0.71 mm).

B. 0.013 inch (0.33 mm) thick of the material specified in item A if used in conjunction with an air spacing of one-half that specified in Table 23.4.

C. A material having a thickness less than that specified in items A and B may be used provided that it is mechanically and thermally equivalent to items A and B and that the material thickness is not less than 5000 volts for item A and 2500 volts for item B.

Exception No. 1: Where the spacing specified in Table 23.4 is provided, the insulation thickness described in items A—C need not be provided.

Exception No. 2: Any type and thickness of insulation in addition to the magnet wire coating, or a through air spacing less than that specified in paragraphs 23.1 and 23.2 may be used between a crossover lead and the winding to which it is connected if the construction complies with either of the following:

A. The coil withstands the appropriate dielectric withstand potential described in items A and C of paragraph 42.1. The potential shall be applied between the coil leads with the crossover lead cut at the point where it enters the inner layer or

B. The coil withstands the induced potential described in paragraphs 42.3—42.6. See paragraph 14.9.

Exception No. 3: This requirement does not apply to insulation between a Class 2 secondary crossover lead and (1) the secondary winding to which the crossover lead is connected, (2) the metallic enclosure, and (3) the core.

14.9 With reference to Exception No. 2 to paragraph 14.8, the magnet coil of a molded-bobbin transformer having a slot for the crossover or start lead — unspliced at the windings — is an acceptable crossover-lead insulation if: (1) the slot provides a graduated through air spacing to the winding, increasing to the end turns, and (2) the magnet-coil winding withstands the induced-potential test described in paragraphs 42.3—42.6.

14.10 Insulating material, such as outer-wrap and crossover-lead insulation, used to reduce the likelihood of live parts becoming accessible through openings in the outer enclosure shall not be less than 0.028 inch (0.71 mm) thick. See paragraph 6.1.

Exception: Insulation need not be provided if the transformer is inaccessible.

14.11 A flanged bobbin-wound transformer shall be constructed so as to maintain physical separation between the primary and secondary windings. Physical separation may be accomplished by employing a 3-flange bobbin for winding the primary and secondary windings adjacent to each other. As an alternative, a telescoping bobbin construction, each section containing an individual winding, may be used where the primary winding is wound over the secondary winding or the secondary winding over the primary winding. The bobbin insulation material shall comply with the requirements in item B of Table 14.1.

Exception: A 2-flange bobbin having the primary winding wound over the secondary winding or the secondary winding wound over the primary with the primary winding insulated from the secondary winding by means of tape insulation is considered acceptable if (1) the tape insulation complies with the requirement in Table 14.1, (2) the tape insulation provides a continuous overlap on the bobbin flanges, (3) the tests described in paragraphs 50.1—50.15 are continued for 15 days, and (4) the induced potential test described in paragraphs 42.3—42.6 is conducted.

15. Resistors

15.1 A resistor shall be reliably supported and the assembly shall be provided with a means other than friction between surfaces to reduce the likelihood of loosening or rotating.

15.2 An assembly employing lock washers may be considered to comply with paragraph 15.1.

16. Switches and Controllers

16.1 A switch or other control device shall be acceptable for the application and shall have current and voltage ratings not less than those of the circuit that it controls when the battery charger is operated as intended.

Exception: A switch or other control device not having an inductive rating that is connected in a transformer secondary circuit and that complies with the Normal Temperature Test, Section 41 and Overload Test, Section 46.

16.2 A primary-circuit switch that controls an inductive load having a power factor less than 75 percent, such as a transformer or some ballasts, and that does not have an inductive rating, shall be rated not less than twice the full-load current rating of the load, or the switch shall be investigated for the application.

16.3 Unless rated for the application, a switch or other device that controls a motor and is not interlocked so that it will not break the locked-rotor motor current shall be subjected to the overload test required by paragraph 46.1 and described in paragraph 46.2.

16.4 A switch that controls a tungsten-filament lamp shall have a tungsten-filament-lamp current rating not less than the maximum current it will control.

Exception: A switch not having an acceptable tungsten-filament-lamp current rating and rated 3 amperes or more may be used to control a 15-watt or smaller lamp.

17. Overload-Protective Devices

17.1 A protective device, such as a fuse or a circuit breaker, the intended functioning of which requires renewal, replacement, or resetting shall be in a readily accessible location.

Exception: A protective device that would ordinarily be unknown to the user because of its location and omission of reference to the device in the operating instructions, circuit diagrams, and other instructional materials provided with the battery charger.

17.2 A household automotive and nonautomotive battery charger shall be provided with a manually or automatically reset protector or other overload-protective device that complies with the requirements in paragraphs 47.1—47.9.

Exception: A battery charger having Class 2 output characteristics need not comply.

17.3 A thermal cutoff or other device employed to reduce a risk of fire or electric shock due to overheating of a transformer during abnormal operation shall comply with the requirements applicable to such a device in addition to the applicable requirements in this standard — for example, a thermal cutoff shall comply with the applicable requirements in this standard and those in the Standard for Thermal Cutoffs for Use in Electrical Appliances and Components, UL 1020.

18. Fuses and Fuseholders

18.1 A fuse and a fuseholder shall have voltage and current ratings acceptable for the circuit in which they are connected.

18.2 An uninsulated live part of a fuseholder that is capable of causing a risk of electric shock, other than the screw shell or clips, shall not be exposed to contact by a person removing or replacing the fuse.

18.3 If a polarized attachment plug is used, the screw shell of a plug fuseholder and the accessible contact of an extractor fuseholder shall be connected toward the load.

18.4 A fuse that is used for output-lead short-circuit protection shall not be interchangeable with a fuse of a higher ampere rating.

19. Receptacles

19.1 If a battery charger includes one or more attachment-plug receptacles for general use, and if the overcurrent protection of a branch circuit to which the charger will be connected exceeds that acceptable for the receptacle or receptacles, each receptacle circuit shall have individual overcurrent protection of not more than 20 amperes installed as part of the charger. This overcurrent protection shall be of the time-delay type.

19.2 With reference to paragraph 19.1, a fuseholder shall be of Type S construction or Edison-base with a factory-installed nonremovable adapter of Type S construction.

19.3 The grounding member of a 15- or 20-ampere general-use attachment-plug receptacle in a battery charger shall be electrically connected to the grounding means of the battery charger.

20. Lampholders

20.1 The screw shell of a lampholder shall be connected to a conductor that is intended to be connected to the grounded conductor of the power-supply circuit.

20.2 A lampholder shall be designed or installed so that uninsulated live parts, other than a screw shell, will not be exposed to contact by persons removing or replacing the lamp in normal service.

21. Capacitors

21.1 A capacitor connected across the line, such as a capacitor for radio-interference elimination or power-factor correction, shall be housed within an enclosure or container that will protect the plates against mechanical damage and will reduce the risk of emission of flame or molten material resulting from breakdown of the capacitor.

21.2 The container of a capacitor shall be of metal providing strength and protection not less than that of uncoated steel having a minimum thickness of 0.020 inch (0.51 mm).

Exception: The container may be of thinner sheet metal or may be of material other than metal, if the capacitor is mounted inside a battery charger having an enclosure that complies with the requirements in paragraphs 5.1—5.3, 5.10, and 5.11.

22. Printed Wiring

22.1 A printed-wiring board shall comply with the requirements in the Standard for Printed-Wiring Boards, UL 796.

Exception: A printed-wiring board that contains only Class 2 circuits, provided deterioration or breakage of the bond between the conductor and the base material would not result in a risk of fire or electric shock.

22.2 A resistor, capacitor, inductor, or other part that is mounted on a printed-circuit board to form a printed-circuit assembly shall be secured so that it cannot be displaced to cause a risk of fire or electric shock by a force likely to be exerted on it during assembly, normal operation, or servicing of the battery charger.

22.3 Consideration is to be given to a barrier or partition that is part of the battery charger assembly and that provides mechanical protection and electrical insulation for a component connected to a printed-circuit board.

23. Spacings

23.1 Spacings at field-wiring terminals shall not be less than specified in Table 23.1 between:

A. Terminals of opposite polarity, and

B. Terminals and other uninsulated metal parts not always of the same polarity.

TABLE 23.1
MINIMUM ACCEPTABLE SPACINGS
AT FIELD-WIRING TERMINALS

Potential Involved, Volts rms	Minimum Spacing, Inch (mm).	
	Over Surface	Through Air
0—150	1/4 (6.4)	1/4 (6.4)
151—300	3/8 (9.5)	1/4 (6.4)
301—600	1/2 (12.7)	3/8 (9.5)

23.2 Spacings, other than at field-wiring terminals, between live parts of opposite polarity and between live and dead metal parts shall not be less than specified in Table 23.2 or 23.3, whichever applies. If an uninsulated live part is not rigidly secured in position by means other than friction between surfaces, or if a movable dead metal part is in proximity to an uninsulated live part, the construction shall be such that at least the minimum acceptable spacings will be maintained.

Exception No. 1: The spacing requirements in Tables 23.2 and 23.3 do not necessarily apply to the inherent spacings of a component of a battery charger, such as a switch, fuse, or attachment plug. Such spacings are to comply with the requirements for the component in question if they are less than the values specified in Tables 23.2 and 23.3.

Exception No. 2: As provided in paragraph 23.4.

23.3 In a Class 2 secondary circuit that does not present a risk of electric shock, spacings between live parts of opposite polarity and between a live part and dead metal that is grounded are not specified.

23.4 An insulating liner or barrier of material such as vulcanized fiber or thermoplastic employed in lieu of required spacings shall not be less than 0.028 inch (0.71 mm) thick and shall be so located or of such material that it will not be adversely affected by arcing.

Exception No. 1: Vulcanized fiber not less than 0.013 inch (0.33 mm) thick or mica not less than 0.0065 inch (0.165 mm) thick may be used (1) in conjunction with an air spacing of not less than 50 percent of the minimum acceptable through-air spacing, (2) in a single-plate rectifier element of an isolated secondary circuit rated 50 volts rms or less, and (3) between a heat sink and a metal mounting surface, including the enclosure, of an isolated secondary circuit rated 50 volts rms or less.

Exception No. 2: An insulating material having a thickness less than that specified in paragraph 23.4 or Exception No. 1 may be used if, upon investigation, it is found to be acceptable for the application and is equivalent in all respects.

TABLE 23.2
MINIMUM ACCEPTABLE SPACINGS OTHER
THAN AT FIELD-WIRING TERMINALS

Potential Involved, Volts	Minimum Spacing, Inch (mm)	
	Through Air	Over Surface
50 or less	1/16 (1.6)	1/16 (1.6)
51—150	1/8 (3.2)	1/4 (6.4)
151—300	1/4 (6.4)	3/8 (9.5)
301—600	3/8 (9.5)	1/2 (12.7)

TABLE 23.3
MINIMUM ACCEPTABLE SPACINGS BETWEEN
AN UNINSULATED LIVE PART AND A METAL
ENCLOSURE OR OTHER ACCESSIBLE DEAD
METAL PART INCLUDING A FITTING FOR
CONDUIT OR ARMORED CABLE

Potential Involved, Volts	Through Air and Over Surface, Inch (mm)
50 or less	1/16 (1.6)
51—150	1/4 (6.4)
151—600	1/2 (12.7)

TABLE 23.4
SPACINGS WITHIN A TRANSFORMER

Potential Involved, Volts	Minimum Spacing
	Through Air and Over Surface, Inch (mm)
250 or less	3/32 (2.4)
251—600	1/4 (6.4)

23.5 Spacings in a motor shall comply with the spacing requirements in the Standard for Electric Motors, UL 1004.

23.6 Film-coated wire is regarded as an uninsulated live part when spacings are being considered.

24. Grounding

24.1 With reference to paragraph 9.20, in a battery charger intended to be connected to the power supply by a metal-enclosed wiring system such as rigid metal conduit or armored cable, or intended to be connected indoors, by a means other than a metal-enclosed wiring system such as nonmetallic-sheathed cable, all exposed dead metal parts and all dead metal parts inside the enclosure that are exposed to contact (1) during intended operation or adjustment of the battery charger or (2) during any servicing operation, including maintenance and repair, and that may become energized, shall be conductively connected to the grounding terminal or lead.

24.2 The surface of a lead intended for connection of an equipment-grounding conductor shall be green with or without one or more yellow stripes, and no other lead shall be so identified.

24.3 A wire-binding screw intended for the connection of a field-installed equipment-grounding conductor shall have a green-colored head that is hexagonal, slotted, or both. A pressure wire connector intended for connection of such conductor shall be plainly identified such as by being marked "G," "GR," "GND," "Ground," "Grounding," or the like; or by a marking on a wiring diagram provided on the battery charger.

24.4 The wire-binding screw or pressure wire connector referred to in paragraph 24.3 shall be located so that it is unlikely to be removed during normal servicing of the battery charger.

24.5 A terminal solely for connection of a grounding conductor shall secure a conductor of the size acceptable for the application in accordance with the National Electrical Code, ANSI/NFPA 70—1984.

24.6 The grounding conductor of a power-supply cord shall be connected to the grounding blade of a grounding attachment plug and shall be connected to dead metal parts within the frame or enclosure by a screw or by an equivalent means that is not likely to be removed during any servicing operation not involving the power-supply cord. Solder alone shall not be used for securing the grounding conductor.

24.7 If a means for grounding of the charger is provided although not required, it shall comply with the requirements in this section.

24.8 The surface of any insulation on the grounding conductor of a flexible cord shall be green with or without one or more yellow stripes and no other conductor shall be so identified.

24.9 A portable or stationary battery charger of the automotive or commercial type or a type for use outdoors shall have provision for grounding dead metal parts in the form of a grounding conductor in the power-supply cord and a grounding attachment plug. See paragraph 9.21.

24.10 A portable or stationary battery charger of the nonautomotive household type intended for operation on a circuit involving a potential of more than 150 volts to ground shall have provision for grounding all exposed dead metal parts, and all dead metal parts that may be exposed or energized during any servicing operation, including maintenance and repair, as specified in paragraphs 24.6—24.8.

PROTECTION AGAINST INJURY TO PERSONS

25. Scope

25.1 The performance and construction requirements in Sections 25—36 are applicable to battery chargers that are covered by this standard and that may involve a risk of injury to persons in normal operation.

25.2 There are risks of injury to persons inherent in some battery chargers that if completely eliminated would defeat the utility of the battery charger. The requirements in Sections 26—36 are intended to reduce such risks, while retaining the normal function of such a battery charger.

26. General

26.1 If the operation and maintenance of a battery charger by the user involves a risk of injury to persons, means shall be provided to reduce the risk.

26.2 When judging a product with respect to the requirement in paragraph 26.1, consideration shall be given to reasonably foreseeable misuse of the product.

26.3 Among the factors to be considered in judging the acceptability of an exposed moving part are:

- A. The degree of exposure necessary to perform its intended function;
- B. The sharpness of the moving part;
- C. The likelihood of unintentional contact;
- D. The speed of the moving part; and
- E. The likelihood that a part of the body would be endangered or that clothing would be entangled by the moving part. These factors are to be considered with respect to both intended operation of the battery charger and reasonably foreseeable misuse.

26.4 Whether a guard, a release, an interlock, or the like is required, and whether such a device is adequate shall be determined from an investigation of the complete battery charger, its operating characteristics, and the likelihood of a risk of injury to persons resulting from a cause other than gross negligence. The investigation shall include consideration of the results of breakdown or malfunction of any component; but not more than one component at a time, unless one event contributes to another. If the investigation shows that breakdown or malfunction of a particular component can result in a risk of injury to persons, that component is to be investigated for reliability.

27. Enclosures and Guards

27.1 A part that could cause injury to persons shall be enclosed.

27.2 An opening in a guard or enclosure around a moving part that may involve a risk of injury to persons shall have a minor dimension less than 1 inch (25.4 mm), and shall not permit the probe illustrated in Figure 6.1 to contact the part when the probe is inserted with a force of 1 pound (4.4 N) through the opening to its maximum depth in a straight or articulated position.

27.3 An enclosure, an opening, a frame, a guard, a knob, a handle, or the like shall not be sufficiently sharp to cause a risk of injury to persons in normal maintenance or use.

27.4 A guard or portion of an enclosure acting as a guard for a part that may involve a risk of injury to persons shall be either (1) mounted to the assembly so that the part cannot be operated with the guard or portion of the enclosure removed, (2) secured to the assembly using fasteners requiring a tool for removal, or (3) provided with an interlock to reduce the risk of contacting the part.

Exception: A commercial battery charger marked as required by paragraph 59.9 need not comply.

28. Enclosure Mounting

28.1 Mounting instructions shall be furnished with a battery charger intended for permanent mounting. If the mounting hardware is not readily available commercially, the manufacturer shall provide the hardware with the charger.

29. Materials

29.1 The material of a part, such as an enclosure, a frame, a guard, or the like, the breakage of which may result in a risk of injury to persons, shall have such properties as to meet the demand of expected loading conditions.

29.2 The requirement in paragraph 29.1 applies to those portions of a part adjacent to a moving part considered to involve a risk of injury to persons.

30. Moving Parts

30.1 A rotating member, such as a fan blade, breakage of which may result in a risk of injury to persons, shall be enclosed or guarded to reduce the risk of injury.

30.2 A rotating or moving part that, if it should become disengaged may create a risk of injury to persons, shall be provided with a means to retain the part in place under conditions of use.

31. Impact

31.1 A battery charger employing a part as described in paragraph 29.1 shall be tested as described in paragraph 31.2 or 31.4; and, following the test, shall:

A. Not permit a probe, as illustrated in Figure 6.1 and applied in accordance with paragraph 27.2, to contact an uninsulated live part or a moving part that may involve a risk of injury to persons; and

B. Comply with the dielectric voltage-withstand requirements in Section 42 with the potential applied between live parts and accessible dead metal parts.

31.2 Each of three samples of a battery charger other than as mentioned in paragraph 31.4 is to be dropped three times from a height of 3 feet (0.9 m) to strike a surface in the positions most likely to produce adverse results. A concrete surface shall be used for an automotive battery charger; all other battery chargers shall be dropped onto a hardwood surface.

31.3 With reference to paragraph 31.2, the hardwood surface is to consist of a layer of nominal 1-inch-thick tongue-and-groove oak flooring mounted on two layers of 3/4-inch-thick plywood. This assembly is to rest on a concrete floor or an equivalent support during the test.

31.4 A battery charger weighing more than 40 pounds (18 kg), a cart-type battery charger, or a battery charger secured to a fixed operating position is to be subjected to a 5-foot-pound (6.8 J) impact. This impact is to be produced by dropping a steel sphere 2 inches (50.8 mm) in diameter and weighing 1.18 pounds (0.54 kg) from a height of 51 inches (1.29 m) to produce the 5-foot-pound impact. For a part not able to be struck from above by the free-falling sphere, the sphere is to be suspended by a cord and allowed to fall as a pendulum through the required vertical distance.

32. Stability

32.1 Under conditions of normal use, a portable or stationary battery charger intended for floor or table use shall not become physically unstable to the degree that it may pose a risk of injury to persons.

Exception: A direct plug-in or cord-connected battery charger weighing 1 pound (0.5 kg) or less.

32.2 A battery charger shall not tip over but shall return to its normal at-rest position when:

- A. Tipped through an angle of 10 degrees from an at-rest position on a horizontal surface,
- B. Placed on a plane inclined at an angle of 10 degrees from the horizontal, or
- C. Subjected to an externally-applied horizontal force of 20 percent of the weight of the battery charger or 50 pounds (22.7 kg), whichever is less. See paragraph 32.5.

32.3 A battery charger is not to be energized during the stability test. The test is to be conducted under conditions most likely to cause the battery charger to overturn. The following conditions are to be such as to result in the least stability:

- A. The position of all adjustable or movable parts such as doors, drawers, or casters;
- B. The supply and output cords placed in either the stored position or on the surface supporting the battery charger, whichever is more severe;

C. Provision of or omission of any normal mechanical load in the battery charger such as stored parts; and

D. Direction in which the battery charger is tipped, or the supporting surface is inclined.

32.4 With reference to the requirement in item A of paragraph 32.2, for a battery charger that is constructed so that while being tipped through an angle of 10 degrees a part or surface of the battery charger not normally in contact with the horizontal supporting surface touches the supporting surface before the charger has been tipped through an angle of 10 degrees, the tipping is to be continued until the surface or plane of the surface of the charger originally in contact with the horizontal supporting surface is at an angle of 10 degrees from the horizontal supporting surface.

32.5 With reference to the requirements in paragraph 32.2, the force specified in item C is to be applied in a horizontal direction at that point on the charger most likely to overturn the battery charger, but is not to be applied more than 5 feet (1.52 m) above floor level. The legs or points of support may be blocked to prevent the unit from sliding during the application of the force.

33. Switches and Controls

33.1 If unintentional operation of a switch can result in a risk of injury to persons, the actuator of the switch shall be located or guarded so that such operation is unlikely.

Exception: A voltage selection switch that complies with the requirements for the abnormal test in paragraph 47.9 need not comply.

33.2 The actuator of a switch may be guarded by recessing, ribs, barriers, or the like.

34. Static Load

34.1 When mounted as recommended by the manufacturer, a battery charger intended to be fixed to a supporting structure shall be loaded as described in paragraph 34.2 for 1 minute with a force equal to three times the weight of the device but not less than 20 pounds (89 N). As a result of this loading, there shall be no permanent deformation, breakage, dislocation, cracking, or other damage to the mounting means.

Exception No. 1: A battery charger weighing 1 pound (0.45 kg) or less need not be tested.

Exception No. 2: A battery charger weighing 20 pounds (9.07 kg) or less and having (1) keyhole slots in a sheet metal enclosure or (2) brackets welded to the enclosure, provided that the enclosure complies with the thickness requirements specified in paragraph 5.3.

34.2 The force is to be applied through the approximate center of gravity of the device, is to be increased gradually so as to reach the required value in 5 to 10 seconds, and is to be maintained at that value for 1 minute.

35. Strength of Handles

35.1 A handle used to lift or carry a charger shall withstand a force of four times the weight of the charger without breakage of the handle, its securing means, or that portion of the enclosure to which the handle is attached.

Exception: A handle used for pushing or pulling a battery charger employing wheels, casters, or the like need not be tested.

35.2 To determine whether a battery charger complies with the requirements in paragraph 35.1, a force equal to four times the weight of the charger is to be uniformly applied over a 3-inch (76-mm) width at the center of the handle, without clamping. The load is to be started at zero and is to be gradually increased so that the test value will be attained in 5 to 10 seconds and maintained at that value for 1 minute. If more than one handle is provided on a charger and the charger cannot be carried by one handle alone, the force is to be distributed between the handles. The distribution of forces is to be determined by measuring the percentage of the weight of the charger sustained by each handle with the charger in the normal carrying position. If a charger is furnished with more than one handle and can be carried by one handle only, each handle shall withstand the total force.

36. Surface Temperatures

36.1 During the temperature test described in Section 41, the temperature of a surface that may be contacted shall not be more than the value specified in Table 36.1. If the test is conducted at a room temperature of other than 25°C (77°F), the results are to be corrected to 25°C.

Exception: A commercial battery charger may exceed the temperature limits specified in the third item in Table 36.1 if the charger is:

- A. Intended to be permanently installed so that the area or surface of the battery charger exceeding the maximum acceptable surface temperatures specified in Table 36.1 is not likely to be contacted by people,
- B. Marked as required by paragraph 59.26, and
- C. Provided with instructions as specified in item G of paragraph 61.5.

TABLE 36.1
MAXIMUM ACCEPTABLE SURFACE TEMPERATURES

Location of Type of Surface	Composition of Surface ^a	
	Metallic	Nonmetallic
	Degrees C F	Degrees C F
Handle or knob for lifting, carrying or holding	50 122	60 140
Handle or knob contacted, but not involving lifting, carrying, or holding; other surfaces subjected to contact in operation	60 140	85 185
A surface subject to casual contact	70 158	95 203

^aA material other than metal that is plated or clad with metal having a thickness of 0.005 inch (0.13 mm) or less is to be judged as a nonmetallic part.

PERFORMANCE

37. General

37.1 A representative sample of a battery charger shall be subjected to the tests described in Sections 38—51. Unless otherwise specified, all tests are to be conducted at the voltage specified in Table 37.1, and at rated frequency. A battery charger rated 50—60 hertz shall be tested at 60 hertz.

TABLE 37.1
VALUES OF TEST VOLTAGES

Rated Voltage rms	Test Voltages rms
110—120	120
Between 121—219	Rated voltage
220—240	240
Between 241—253	Rated voltage
254—277	277
Between 278—439	Rated voltage
440—480	480
Between 481—525	Rated voltage
550—600	600

38. Leakage Current Test

38.1 The leakage current of a cord-connected battery charger, when tested in accordance with paragraphs 38.2—38.7, shall not be more than:

A. 0.5 milliamperere for a portable charger, or

B. 0.75 milliamperere for a stationary or fixed battery charger employing a standard attachment plug rated 20 amperes or less.

38.2 All exposed conductive surfaces are to be tested for leakage currents. The leakage currents from these surfaces are to be measured to the grounded supply conductor individually as well as collectively if simultaneously accessible, and from one surface to another if simultaneously accessible. Parts are considered to be exposed surfaces unless guarded by an enclosure considered acceptable for protection to reduce the risk of electric shock as defined in paragraph 6.1. Surfaces are considered to be simultaneously accessible if they can be readily contacted by one or both hands of a person at the same time. These measurements do not apply to output terminals operating at voltages less than 30 volts rms — 42.4 volts peak.

38.3 If a conductive surface other than metal is used for the enclosure or part of the enclosure, the leakage current is to be measured using a metal foil with an area of 10 by 20 centimeters in contact with the surface. If the surface is less than 10 by 20 centimeters, the metal foil is to be the same size as the surface. The metal foil is not to remain in place long enough to affect the temperature of the battery charger.

38.4 The circuit for the leakage current measurement is to be as illustrated in Figure 38.1. The measurement instrument is defined in items A—C. The meter that is actually used for a measurement need only indicate the same numerical value for a particular measurement as would the defined instrument; it need not have all the attributes of the defined instrument.

A. The meter is to have an input impedance of 1500 ohms resistive shunted by a capacitance of 0.15 microfarad.

B. The meter is to indicate 1.11 times the average of the full-wave rectified composite waveform of voltage across the resistor or current through the resistor.

C. Over a frequency range of 0—100 kilohertz, the measurement circuit is to have a frequency response — ratio of indicated to actual value of current — that is equal to the ratio of the impedance of a 1500-ohm resistor shunted by a 0.15-microfarad capacitor to 1500 ohms. At an indication of 0.5 or 0.75 milliamperes, the measurement is not to have an error of more than 5 percent at 60 hertz.

38.5 Unless the meter is being used to measure leakage current from one part of a battery charger to another, the meter is to be connected between an accessible part and the grounded supply conductor.

38.6 A sample of the battery charger is to be tested for leakage current starting with the as-received condition — the as-received condition is to be without prior energization except as may occur as part of the production-line testing — but with the grounding conductor, if any, open at the attachment plug. The supply voltage is to be adjusted to the voltage specified in Table 37.1. The test sequence, with reference to the measuring circuit, Figure 38.1, is to be as follows:

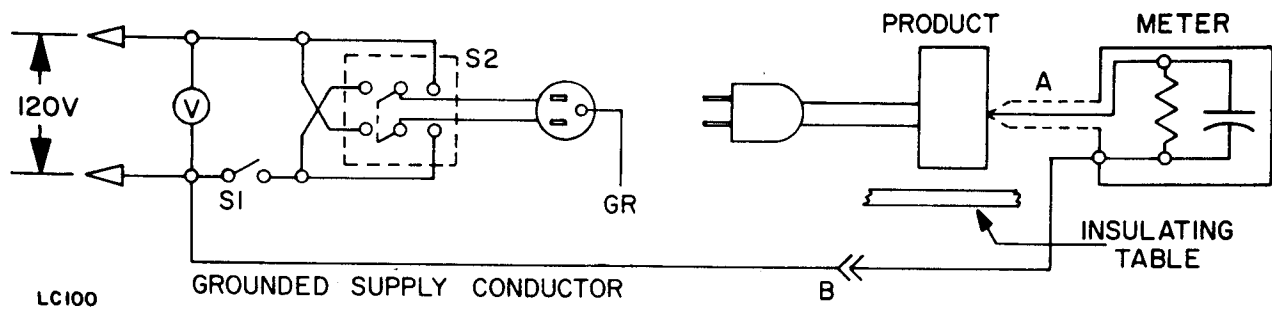
A. With switch S1 open, the battery charger is to be without load and connected to the measuring circuit. Leakage current is to be measured using both positions of switch S2 and with the battery charger switching devices in all their intended operating positions.

B. Switch S1 is then to be closed energizing the battery charger, and within 5 seconds, the leakage current is to be measured using both positions of switch S2 and with the battery charger switching devices in all their intended operating positions.

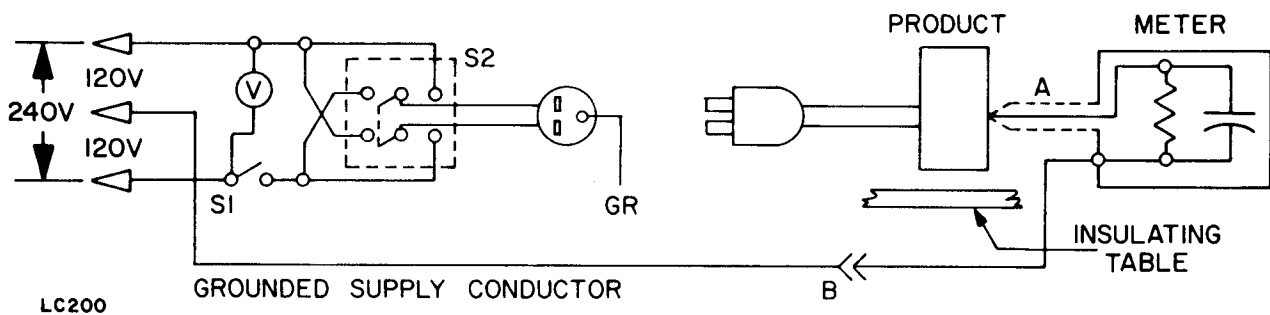
C. Leakage current is to be monitored until thermal stabilization. Both positions of switch S2 are to be used in making this measurement. Thermal stabilization is considered to be obtained by operation of the battery charger as in the normal temperature test.

38.7 Normally, the complete leakage current test, as described in paragraph 38.6, is to be conducted without interruption for other tests. With the concurrence of those concerned, the leakage current test may be interrupted to conduct other nondestructive tests.

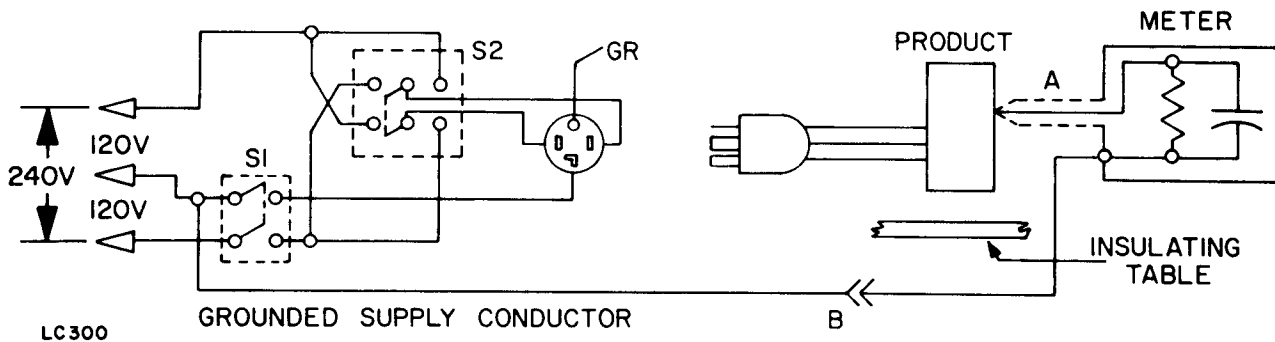
FIGURE 38.1
LEAKAGE CURRENT MEASUREMENT CIRCUITS



Appliance intended for connection to a 120 volt power supply.



Appliance intended for connection to a 3-wire, grounded neutral power supply, as illustrated above.



Appliance intended for connection to a 3-wire, grounded neutral power supply, as illustrated above.

NOTE: A — PROBE WITH A SHIELDED LEAD.
B — SEPARATED AND USED AS A CLIP WHEN
MEASURING CURRENTS FROM ONE PART OF A DEVICE
TO ANOTHER

39. Power Input Test

39.1 The input current and power factor to a battery charger shall be measured with the charger operating while connected to the maximum normal load. See paragraphs 39.2—39.5. The current input shall not be more than 110 percent of the rated value.

39.2 The maximum normal load for a battery charger having a rating of 20 amperes or less and intended to charge lead-acid storage batteries is to be a 100,000-microfarad capacitor in parallel with a resistor adjusted to draw rated output current. A battery charger for series charging of more than one battery is to be tested with a resistive-capacitive load adjusted to represent the additional batteries.

39.3 A household, nonautomotive charger for a small rechargeable battery, such as a nickel-cadmium type, is to be tested using the battery for which it is intended as a load.

39.4 A battery charger having an output rating of more than 20 amperes and intended to charge lead-acid storage batteries is to be tested with (1) a resistive-capacitive load having a capacitance based on the battery ampere-hour rating as specified in Table 39.1, (2) a battery supplement with a resistive load bank, or (3) an acceptable battery if requested by the manufacturer. A battery rated 70 ampere-hours or a resistive-capacitive load having a capacitance of 185,000 microfarads is to be considered a representative automotive load. A marine battery charger that is not current-limited is to be tested with an acceptable battery.

TABLE 39.1
CAPACITANCE

Maximum Rated Battery Amperes-Hours	Measured Battery Capacitance, Farads	Capacitance ^a Farads/ Ampere-Hour
10	0.16	0.016
50	1.05	0.021
300	5.1	0.017
840	13.5	0.016

^aPer cell at full charge.

39.5 If a charger is to be tested using a battery or batteries as the load, the battery is to be discharged to 1.75 volts per cell at a rate not to exceed the discharge rate assigned by the battery manufacturer, but in any case, the rate of discharge is not to exceed one-sixth of the ampere-hour capacity of the battery.

Exception: The discharge rate of not more than one-sixth of the ampere-hour capacity of the battery does not apply to a specific purpose battery charger; for example, a household nonautomotive charger used to charge a small rechargeable battery, such as a nickel-cadmium battery.

39.6 A battery charger having an engine-cranking-circuit switch shall also comply with paragraph 39.1 with the switch in the cranking position.

39.7 With reference to paragraph 39.1, for an automotive battery charger provided with a tap switch or other means for adjustment of the output current or voltage, measurements of the primary input current and the secondary output voltage shall be made with the output load adjusted to the normal rated output current at each tap-switch position. For the temperature test the switch shall be set at the tap position resulting in the output voltage nearest the value specified in Table 39.2 for the output-current range that includes the output current rating marked on the charger.

39.8 The output voltage of a battery charger intended to charge a series of either 6- or 12-volt batteries is to be a direct multiple of the voltage for the corresponding current specified in Table 39.2.

TABLE 39.2
OUTPUT VOLTAGE

Rated Output Current, Amperes	Closed Circuit Output Voltages ^a	
	6-Volt Battery	12-Volt Battery
0—4.99	6.4	12.8
5—14.9	6.7	13.4
15—49.9	7.1	14.2
50—79.9	7.3	14.6
80—100	7.4	14.8

^aClosed-circuit voltage is the voltage measured with the charger connected to a battery that has been discharged in accordance with paragraph 39.5 and without a load on the battery.

39.9 Output current measurements of either half- or full-wave rectified circuits are to be based on the average current.

40. Output Voltage Measurement Test

40.1 The output voltage of an automotive battery charger having an output current rating of 20 amperes or less shall be at least 90 percent of the value specified in Table 39.2. An average-reading meter is to be used for measuring the output.

40.2 To determine whether a battery charger complies with the requirements in paragraph 40.1, a variable resistive-capacitance load is to be connected to the output terminals. The battery charger is to be at room temperature at the beginning the the test, and output measurements are to be taken approximately 5 minutes after energization of the primary, using an average reading meter.

41. Normal Temperature Test

General

41.1 A battery charger shall be tested as described in paragraphs 41.2—41.14, using a load adjusted to maximum normal load current as described in paragraphs 39.2—39.5, and shall not reach a temperature at any point high enough to cause a risk of fire, to damage any material used, or to exceed the temperature rises specified in Table 41.2.

41.2 In the temperature test of a battery charger using a battery load, the battery shall be discharged as specified in paragraph 39.5, then charged until temperatures on the charger reach a maximum and begin to decrease. The temperature test is to be continued using a second battery, also discharged as specified in paragraph 39.5, until maximum temperatures are attained.

41.3 An automatic protector or other recycling device is not to operate during the normal temperature test.

41.4 A battery charger designed for mounting or support in more than one position, or in a confined location, shall be tested in a manner representing the most severe conditions. An adjacent mounting or supporting surface shall consist of nominal 1-inch thick, soft-pine boards.

41.5 Unless investigated and found acceptable — see paragraph 5.9 — a supporting means formed of soft rubber or a rubber-like material is to be removed prior to the temperature test. If the supporting means has a metal insert, such as a screw or rivet, the test is to be conducted with the battery charger supported by the metal insert. At the request of the manufacturer, the test may be conducted without any means of support.

Discharge Temperature Test

41.6 A battery charger employing a discharge resistor for battery testing shall be subjected to a discharge temperature test. The temperatures measured during the first two discharges shall not exceed the temperature rises specified in Table 41.2. There shall be no emission of flame or molten metal from the charger during any of the five discharges.

41.7 Starting within 1 minute after completion of the temperature test described in paragraph 41.1, a full-charged battery of the required capacity is to be discharged through the discharge circuit. The length of the discharge time is to be as specified in Table 41.1. The discharge is to be repeated at 2-minute intervals until a total of five discharges have occurred.

TABLE 41.1
DISCHARGE TEST TIME

Length of Discharge Recommended by Manufacturer, Minutes	Discharge Test Time
2 or less	150 percent of time recommended by manufacturer
More than 2	125 percent of time recommended by manufacturer
No recommendation	1-1/2 minutes